

**AFRL-Rocket Propulsion Division, Aerospace Systems Directorate
Edwards AFB**

Attachment 1: “Questions and Answers”

For

“Space Propulsion Research in Neutralizing Threats”

BAA-RQR-2019-0001

Date: 8 April 2019

This attachment including the cover page consists of 14 pages

Questions and Answers (4 Apr 2018):

Q: What specific technologies within the topic areas are you seeking information for?

A: Per the BAA, Section I-1 "Statement of Needs" lists six topic areas for this BAA. The purpose of this BAA is for any technology that relates to one or more of those topic areas.

For informational purposes including the current state of research and goals of the AFRL RQRS branch, please see Attachment 1: "The Air Force Research Laboratory's In-Space Propulsion Program" (Distribution Statement A: PA Clearance Number 19059).

Attachments:

The Air Force Research Laboratory's In-Space Propulsion Program (Distribution Statement A: PA Clearance Number 19059).

The Air Force Research Laboratory's In-Space Propulsion Program¹

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Abstract

The Air Force Research Laboratory (AFRL) performs R&D for development and sustainment of advanced space propulsion technology for DoD applications. Core programs span a broad range of propulsion options including both chemical and electric propulsion technologies. This presentation will provide a brief survey of the four main space propulsion technology areas under investigation at AFRL. The Test/Flight demonstration area primarily supports legacy systems and enhances the transition/integration of existing propulsion technologies, including HETs, onto DoD spacecraft. The advanced chemical propulsion area is focused on the replacement of hydrazine in the DoD fleet with higher performance IL formulations. Finally, the electrospray and pulsed electromagnetic areas are investigating the practical performance potential for high efficiency electric propulsion technologies.

Introduction

The Air Force Research Laboratory (AFRL) is the technology development arm of the United States Air Force (USAF). Its mission is to “lead the discovery, development and integration of affordable warfighting technologies for our air, space, and cyberspace force.” AFRL consists of approximately 10,000 total employees, including civil servants (45%), uniformed military members (15%), and support contractors (40%). Of the >6000 AFRL members who are scientists or engineers, 80% have at least a Master’s degree with approximately 1/3 possessing a doctorate in their area of expertise. AFRL personnel and activities are dispersed among the locations shown in Figure 1.

AFRL’s Rocket Propulsion Division (AFRL/RQR) located at Edwards Air Force Base, is responsible for research, development, and maturation of liquid rocket engine, solid rocket motor and spacecraft propulsion technologies for use on future Air Force and Department of Defense (DoD) spacecraft. Organizationally, it fits within the AFRL Aerospace Systems Directorate (AFRL/RQ) headquartered at Wright Patterson Air Force Base in Dayton, Ohio.

¹ Note: This paper is an update of an earlier review paper presented at this conference in Ref [0]

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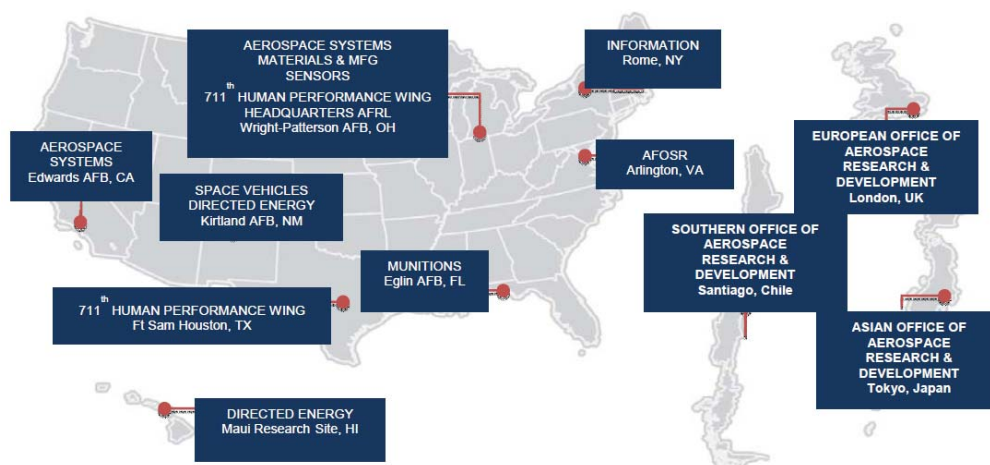


Figure 1. AFRL locations and major offices

1. Background on Spacecraft Propulsion

Fundamentally, all conventional spacecraft propulsion is fundamentally a reaction engine, where a consumable reaction mass (propellant) is exhausted at high speed away from the spacecraft, resulting in an equal and opposite force (thrust) on the spacecraft. The two major types of spacecraft propulsion are chemical thrusters (where the propellant provides both the energy source and the reaction mass) and electric propulsion (EP) thrusters (where the energy source is independent of the propellant). Within the world of high performance EP, both electrostatic (HET and Ion engines) and electromagnetic configurations are used for propellant acceleration. Key characteristics of both types of spacecraft propulsion are summarized in Table 1.

Chemical thrusters	High performance EP
- Propellant chemistry defines available energy for propulsion	- External power source (typically solar panels) define available energy for propulsion
- Lifetimes measured in 10s hours to 10s of minutes	- Lifetime measure in thousands of hours
- Hydrodynamic nozzle used to efficiently extract thrust from high enthalpy combustion products	- Requires formation of charged species (typically plasma) to allow for direct energy transfer to propellant acceleration (relatively low fraction into thermal energy of propellant)
- Low amu products, high flame temp, storable liquid propellants (ex. N_2H_2 , N_2O_4)	- High amu, low ionization energy, storable high pressure gaseous propellants (ex. Xe, Ne)
- Average propellant exhaust speed 2-3 km/s	- Average propellant exhaust speeds >15 km/s
- Scalable from 0.1N to >100lbf thrust	- Scalable from <1 nN to >1 N thrust
- Lifetime of >1-10 hours	- Lifetime of >1000 hours
- Approximate as impulsive burns	- Characteristic spiral orbits

Table 1. Characteristics of conventional space propulsion systems

2. Technical Goal Development

AFRL's spacecraft propulsion technology development goals are derived from two primary sources. The first is the AFSPC Core Function Support Plan (CFSP), which provides high-level guidance regarding capability advances that are required in both specific components (e.g. transceivers, detectors, etc.) and pervasive capabilities (e.g. thermal control, propulsion, etc.). Using CFSP guidance, a team of AFRL and Space & Missile Systems Center (SMC) personnel work together to maintain an SMC "Tech Needs" document that establishes quantitative performance targets for individual systems and subsystems.

The second source of technology goals is the Rocket Propulsion for the 21st Century (RP21) program, which is the successor to the Integrated High Payoff Rocket Propulsion Technologies (IHPRT) effort. RP21 is a government-industry consortium that is co-chaired by senior Department of Defense and NASA officials, who are supported by a steering committee consisting of industry representatives. Its purpose is to coordinate mutually beneficial rocket propulsion investment across both the defense and civil sectors of the government and industry base. The program operates on a philosophy of "technology push," meaning that it is intended to advance the overall state of U.S. rocket propulsion capabilities, rather than to develop point designs that are optimized for a particular launch vehicle or spacecraft. This is facilitated by choosing a baseline system representing a given technology area and then specifying quantitative improvements that are desired in certain metrics and timeframes. Relevant RP21 goals are shown in Table 2. (Note: These goals are revised at regular intervals and are presently under revision.)

	2017	2027
Increase Efficiency (ET/ES/EM)	15%/15%/10%	65%35%/30%
Decrease EP System Dry Mass (ET/ES/EM)	0%/50%/50%	75%/90%/90%
Decrease Flexible Prop.System Wet Mass	35%	65%
Increase Chemical Prop. Density Isp	5%	15%
Decrease Chemical Prop. Dry Mass	10%	40%

Table 2. Recent RP21 Spacecraft Propulsion Goals

While the SMC Tech Needs and RP21 goals are separate products, there is substantial overlap between the two. AFRL plays a fairly large role in development of both. One of the primary ways in which AFRL contributes is by performing rudimentary mission analyses to determine which propulsion parameter improvements would provide the greatest payoff at the mission level for certain classes of spacecraft, as opposed to an individual spacecraft design. For instance, a majority of USAF spacecraft operate in geosynchronous earth orbit (GEO). A common mission scenario that is often considered is therefore the transfer of a spacecraft from an initial injection orbit to GEO using electric propulsion, as performed by the AEHF-1 spacecraft. [2] [3]. The fine details required to conduct such a transfer are fairly complicated and depend on many parameters including launch date, spacecraft steering and attitude control algorithms, and similar design-specific nuances. The general mission performance and trends, however, can be readily studied using two simple relations given in Equations (1) and (2). Here η represents the efficiency of the propulsion system, g is the gravitational acceleration (9.81 m/s^2), I_{sp} is specific impulse (for rockets, I_{sp} is defined as the propellant exhaust velocity divided by g), and T/P is the thrust-to-power ratio. The delivered payload mass fraction is represented by m_{pay}/m_o , α is the specific power of the combined power and propulsion subsystems (W/kg), ΔV is the velocity increment associated with a given orbit change, and ΔT_{trip} is the allotted thruster firing time to complete the transfer. In this formulation, the delivered spacecraft mass, m_{pay} , is defined as the spacecraft mass delivered to the destination orbit less the dry mass of the propulsion and power subsystems.

$$\eta = \frac{gI_{sp}}{2} (T/p) \quad (1)$$

$$\frac{m_{pay}}{m_o} = \frac{(gI_{sp})^2 + 2\eta\alpha\Delta T_{trip}}{2\eta\alpha\Delta T_{trip}} e^{-\Delta V/gI_{sp}} - \frac{(gI_{sp})^2}{2\eta\alpha\Delta T_{trip}} \quad (2)$$

A simple analysis of a 90-day low Earth orbit (LEO) to GEO transfer using only Equations (1) and (2) is shown in Figure 2 for an allotted thruster firing time of 90 days and several combinations of the electric propulsion (EP) system efficiency, η , and specific mass, α . This figure shows several mission-level performance benefits that can be obtained by improving both from typical state of the art values of roughly $\alpha=50$ W/kg and $\eta=60\%$. Also, this analysis indicates that the optimum thruster specific impulse assuming current and near-term propulsion technology is in the 1500-3000 second range. Beyond this analysis, longer allowable trip times tend to favor higher specific impulses, while shorter trips, more energetic starting orbits, and practical limits on spacecraft power level for specific missions (e.g. due to cost, size, etc.) tend to reduce the optimum specific impulse.

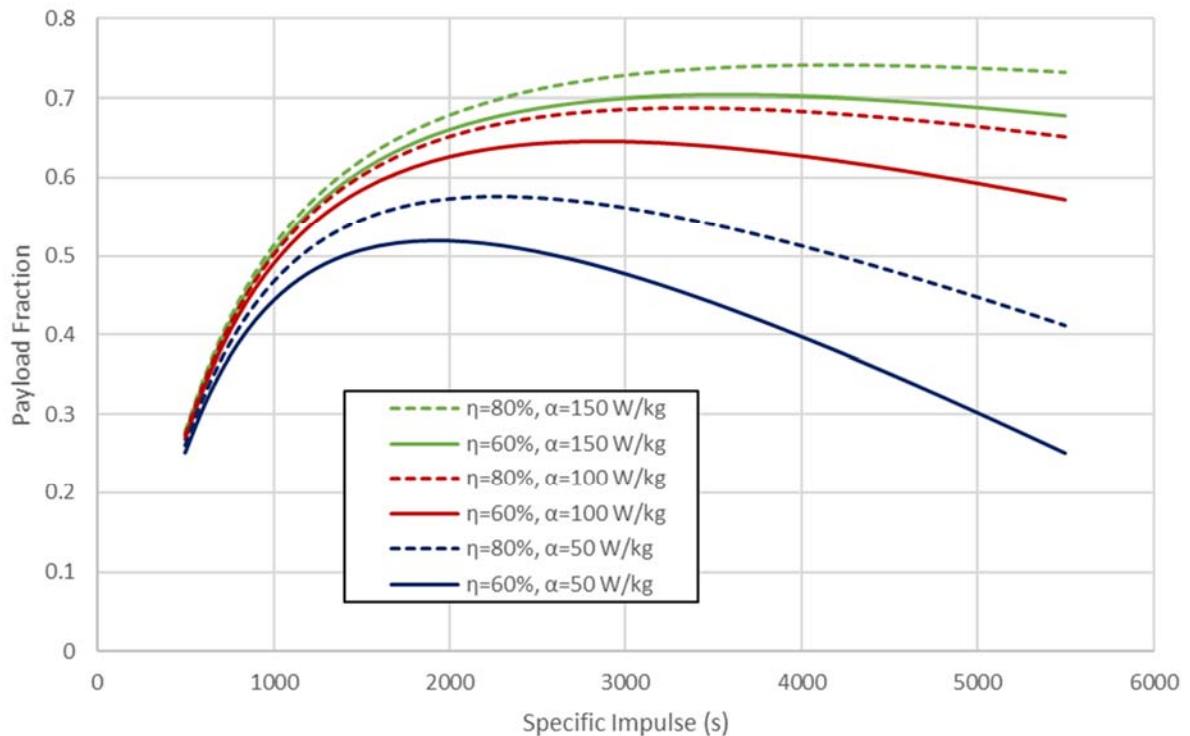


Figure 2. 90 Day LEO-GEO constant thrust transfer

A family of analyses like the one described above, as well as analogous assessments for chemical propulsion and other applications, are used to aid the formation of the specific technology goals pursued by AFRL/RQR. However, despite the effort that goes into the analyses, there are both many second and third order effects which are not captured and also intangible “-ilities” which are difficult to quantify for the many potential scenarios of future spacecraft operations. As an example, a common thread between most of the advanced thruster concepts under exploration is the flexibility goal to run on a common propellant, allowing for operation at high thrust and at high efficiency thruster on a single propellant (multimode propulsion).

3. Technology Portfolio

To address the technology goals introduced in the previous section, the In-Space Propulsion Branch has a number of efforts, both in-house and contracted, and spanning the gamut from basic science research in collaboration with university researchers to applied technology development programs executed by the US industrial base. Moreover, with the very high threshold required for TRL advancement in the space industry, technology transition activities are often accomplished via space experiments or incorporation of AFRL-developed hardware or expertise into spacecraft acquisition programs in partnership with DoD customers. Finally, the In-Space Propulsion Branch maintains an ongoing advisory role in the sustainment of mature propulsion systems. For instance, HET technology is rapidly maturing and has broad acceptance across the commercial marketplace; therefore, while AFRL/RQR no longer has an active program to enhance HET technology, it continues to regularly support DoD customers with aspects of HET testing and integration. These activities can be summarized into four overarching areas of R&D at AFRL as shown in Table 2.

Area	Goals
Test/Flight Demonstration	- Support utilization of existing Advanced EP/Chemical propulsion technology including validation of computational tools for thruster/spacecraft integration - Evaluate thruster behavior / environment impact in vacuum environment
Advanced Ionic Liquid (IL) Chemical Propulsion	- Develop next generation of higher performing IL-based monopropellants to replace existing hydrazine architecture - Explore potential for IL bipropellant formulations
IL Electrospray (ES) propulsion	- Identify performance limits of IL-ES thrusters (thrust density per unit area, efficiency, thruster system dry mass, lifetime) - Identify and formulate improved IL propellant formulations
Pulsed electromagnetic propulsion (primarily Field Reversed Configuration – FRC)	- Understand energy loss mechanism of plasmoid formation at low energy per pulse - Develop tools for designing antenna/capacitor/switch system to optimize formation and translation efficiency

Table 3. AFRL technology portfolio overview (roughly in order from most mature to least mature technology areas)

Test/Flight Demonstration

Given the extreme difficulty in servicing spacecraft, effective transition of space propulsion technology requires extensive test and evaluation (T&E) to reliably quantify the performance and integration characteristics in the space environment. This challenge means that even higher TRL technologies (such as advanced chemical monopropellants and HETs), which are at or near operational status, still require a significant degree of T&E for physical modifications, lifetime extension, and/or performance envelope expansion. AFRL/RQR uses a number of local vacuum facilities for T&E of both EP and chemical systems. In addition, AFRL maintains strong relationships with various agencies for space flight test opportunities. Both ground testing and flight experiments are closely tied to ongoing efforts to validate AFRL-developed plasma plume-spacecraft integration codes [4] for satellite design.

One particularly exciting aspect of T&E is the rapidly evolving capability of high speed sensors/DAQs to provide a detailed picture of the dynamics of HET discharges. While from an operator's perspective, a

HETs can accurately be assumed to function as an essentially steady-state device, the dynamics of discharge actually embed a great deal of basic physics on the operation (and particularly the nonlinear behaviors) of these devices. The EP TEMPEST project (sponsored by AFOSR) is exploring the use of these high speed dynamics to validate physics-based HET models to better understand the details of thruster operation across a range of operating environments (from vacuum chamber to LEO to GEO.) Current efforts center on improved data assimilation procedures to leverage this new test capability and the use of time-delay representations to improve model calibration procedures for HET simulation tools.

Advanced Ionic Liquid (IL) Chemical Propulsion

AFRL's primary activities in the field of spacecraft advanced chemical propulsion are focused on development of hydrazine-replacement technologies. Over the last decade, a class of reduced toxicity, high performance monopropellants consisting primarily of ionic liquids (low melting point salts) has been developed to meet the IHPRT goal of a 50% increase in density-impulse over hydrazine [5]. One of these propellants, known as AF-M315E, was selected for maturation and technology transition based on both its high performance and its very benign safety properties. In particular, AF-M315E exhibits essentially no vapor toxicity, acceptable thermal stability, and is insensitive to electrostatic discharge at spark energies of 1.0 joule [5]. It has been categorized as a slight irritant per the Dermal Irritation Descriptive Classification and a non-irritant according to the European Dermal Evaluation Criteria for erythema and edema as opposed to hydrazine, which is such a strong irritant that it is categorized as corrosive [5]. Use of AF-M315E is expected to reduce costs and timelines for launch site operations because it can be handled in minimal protective equipment (e.g. shirtsleeves) rather than requiring full SCAPE suits and breathing apparatus.

After several years of steady progress in development of thrusters capable of operating on AFM315E, and withstanding the intense chamber conditions that enable high-performance, thrusters using this propellant have been selected for flight demonstration as the highlight of NASA's Green Propellant Infusion Mission (GPIM) [6] with projected launch in 2019. GPIM will demonstrate thrusters operating at the 1N thrust levels [7]. Ball Aerospace is the prime contractor for GPIM, Aerojet-Rocketdyne is building the propulsion system, and AFRL is furnishing the AF-M315E propellant as well as performing the propellant loading operations. AFRL is also pursuing a technology development program to advance current AF-M315E technology by developing and demonstrating thruster hardware capable of providing similar lifetimes and duty-cycles when compared to hydrazine with flight-ready thruster demonstration at the 1N and 22N thrust levels.

In addition to these efforts to transition AF-M315E to operational use, AFRL/RQR work in advanced chemical propulsion also includes ongoing exploration of new propellant formulations, creation of improved thruster test facilities and procedures, and adaptation of monopropellant devices to multiple mission classes (very small impulse bits, high thrust scaleup, etc). These activities are supported by a growing modeling and simulation (M&S) component. Leveraging the comprehensive test capabilities available at AFRL, the current M&S effort focus is on model calibration of system and 1D models for monopropellant behavior rather than high dimensional, first principles-based physics modeling; however, significant efforts performed in conjunction with the Propellants Branch at Edwards (AFRL/RQRP), are using considerably more detailed physics-based approaches (quantum chemistry and molecular dynamics) to accelerate the development of effective chemical kinetics models for IL propellants.

IL Electrospray (ES) propulsion

All plasma-based EP systems, including HETs, have to expend significant energy to separate electrons from neutral propellant atoms. By contrast, the fundamental charged nature of the anion and cation in ILs also enables operation of IL-ES thrusters without the traditional energy losses to propellant ionization, leading to very high efficiencies (up to 80% [8]). This high efficiency is particularly desirable for small thrusters since the large surface area to volume ratio (corresponding to the potential energy loss divided by the region for plasma generation) for small EP thrusters do not display the same high performance characteristics as larger HET/Ion engines³.

In recent years, AFRL has ramped up programs to investigate the potential of IL-based electrospray (ES) technology as a viable path towards high T/P EP thruster systems with initial application to small spacecraft. A major push of this program is to investigate the feasibility of thrust densification for this technology – while it has been successfully space demonstrated on very small scales with the LISA pathfinder mission [10], realistic application of this technology for primary propulsion requires order of magnitude improved thrust density. Given the theoretical performance projected for emission from highly multiplexed electrospray tips [11], illustrated in Fig. 3, research at RQR has focused on understanding the practical limitations of pure ionic regime (PIR) ES operation. In addition, AFRL has a parallel efforts to investigate both the operation of ES on the same energetic ionic liquid propellants used for advanced chemical propulsion (Multimode Propulsion), as well as exploring new IL formulations geared solely towards even higher performance ES operation. Future activities include investigation of ES thruster design and operation to allow for control of transition from ionic (1500-3000s) to cone-jet mode (<800s) for truly high T/P throttleable EP propulsion.

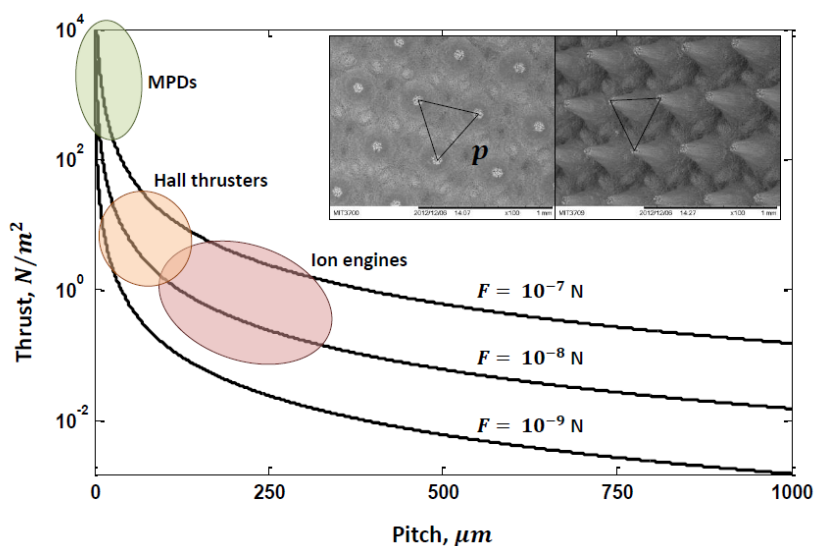


Figure 3. Scaling of thrust density versus electrospray tip pitch from [11]

³ The excellent performance of HET thrusters at the <1 kW – 10+ kW level (i.e. efficiencies >60% over Isp range of 1500-3000s), coupled with their rapidly growing flight heritage, particularly in commercial systems, makes these devices the current standard for high efficiency EP. Ion thrusters typically operate at even high Isp (3000-4000s) and are at on-par or even greater TRL maturity; however, there remains reduced broadband commercial acceptance versus HETs. For a review of recent EP technologies, see [9]

Pulsed electromagnetic propulsion

In the direction of EP at very high power levels, NASA is exploring the potential for near-term systems at the 200kW power level with nested HET technology [12]. Given the relative technological maturity of HET technology, the propulsion system specific mass targets (<5 kg/kW) provide a credible path towards significantly lighter EP systems with efficiencies comparable to present SoA. AFRL research in the area of very high power EP is centered on even more aggressive long-term goals based on pulsed plasmoid electric propulsion (specifically on the Field Reversed Configuration (FRC) concept) where theoretical efforts and very early engineering efforts have identified thruster efficiencies of $>85\%$ [13] and propulsion system specific mass estimates of 1.5 kg/kW [14]. Besides potential improvements to performances, the acceleration of high density magnetic plasmoids offers two other key features. The first, based on leveraging the large relatively elastic charge exchange cross section for many propellants, allows for neutral entrainment [15] into the ejected plasmoid, effectively providing a highly efficient thruster over an extremely large I_{sp} throttle range. The second is very wide propellant compatibility due to the inductive nature of plasmoid formation and acceleration obviating the need for a electrodes in contact with the plasma environment.

While FRC research has been ongoing for over a decade at AFRL, the challenges of scaling the power level of this technology from fusion application (many kJ/pulse) down to space application (on the order of <1 J/pulse) are immense. Accordingly, the AFRL direction in this area has evolved to focus on the plasmoid formation stage (as opposed to the simpler translation/acceleration stage) of the pulse with a very strong emphasis on code development and validation. A major lesson from the history of FRC hardware development is the difficulty in optimizing the efficiency of a complex dynamical system relative to a quasi-steady system such as an HET – not only are there approximately as many steady-state free parameters in engineering design of the physical system (coil windings, cone angle, flux conserver locations, preionization system, etc), but the temporal evolution of the driving EM pulse [17], as shown in Fig. 3, is a time-varying control parameter (directly coupled to the plasma evolution) affecting the overall performance of the thruster. Therefore, to effectively explore the potential of this technology to exceed HET performance (which relies on steady-state control parameters and has the benefit of 50+ years of experimentation), AFRL has designed a science-focused program to develop better fundamental understanding of high density pulsed plasma behavior as an enabler for future engineering efforts. Experimentally, this program is focused on integrating faster plasma diagnostics (both probe and optical) into the existing laboratory setup. Numerically, it has spurred the development of new M&S core competencies for FRC modeling including multifluid MHD [18] and collisional-radiative modeling [19] with a future emphasis on radiation transport to better understand energy migration during plasmoid formation. Finally, early progress for future design optimization activities has been demonstrated by using machine learning on physics-based simulation results to better understand the nonlinear response of the FRC to control parameters [20].

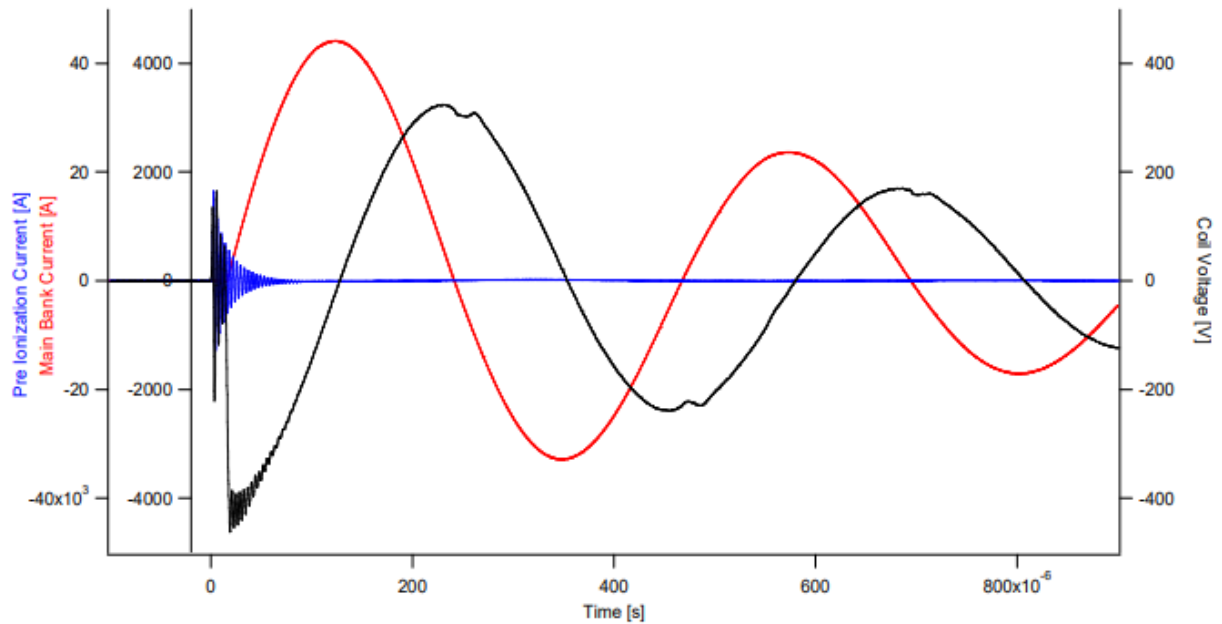


Figure 4. Coil performance for XOCOT-T FRC thruster from Ref [17]

4. Conclusion

The In-Space Propulsion Branch of the Air Force Research Laboratory works closely with partners in the Department of Defense and industry to develop technologies to advance national capabilities in the field of spacecraft propulsion. Significant efforts leverage extensive T&E capabilities (both experimental and flight) and broad involvement in advanced IL monopropellant thrusters to provide near-term technology and support to DoD space assets. Finally, long-term research programs are underway for both electrospray and pulsed electromagnetic thruster technology to better understand potential EP candidates for future multimode propulsion systems.

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